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comfos-nasc2026@math.upd.edu.ph

Analysing Collective Behaviour in Animal Species

Clark Kendrick Go^{*1}

¹ Collaborative Analytics, Department of Mathematics, Ateneo de Manila University

*contact: cgo@ateneo.edu

Collective behaviour in animal groups is the synchronised dynamics of individual agents as a result of their interactions among each other. A variety of mathematical models already exist to describe these interactions and resulting behaviour - with each model catering to different assumptions and motivations [1, 2, 4]. Central to the understanding of collective behaviour are these questions: how does information transfer from one agent to another, and how are these measured? How do seemingly random movements result in emergent behaviour? How do information-theoretic techniques reveal leadership in animal groups?

In this talk, we explore a number of techniques where interactions among agents are studied. First, we talk about general models for agent interaction using self-propelled particle models like the D'Orsogna models [3, 5]. Next, we explore use cases in different animal species when time series data are employed. In both models, we explore numerical techniques that analyse animal interactions and their corresponding emergent behaviour.

Keywords: collective behaviour, information theory, transfer entropy

MSC2020: 94A17, 68T99

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